

Geodesic Interior-Point Method for Cones with Log-Homogeneous Barriers

1 Problem

We consider the conic program

$$\min_{x \in \mathbb{R}^p} c^\top x \quad \text{s.t.} \quad Ax + b \in K, \quad (1)$$

where $K \subset \mathbb{R}^m$ is a proper cone with a ν -logarithmically homogeneous self-concordant barrier (LHSCB) $F : \text{int}(K) \rightarrow \mathbb{R}$:

$$F(t \cdot s) = F(s) - \nu \log t, \quad t > 0.$$

This implies the scaling identities

$$\nabla F(ts) = \frac{1}{t} \nabla F(s), \quad \nabla^2 F(ts) = \frac{1}{t^2} \nabla^2 F(s). \quad (2)$$

2 Primal-dual variables

The slack is $s = Ax + b \in \text{int}(K)$. The dual variable is defined via the barrier gradient:

$$\lambda = -\mu \nabla F(s), \quad \mu = \frac{1}{k^2},$$

which satisfies $\lambda \in \text{int}(K^*)$ (since $-\nabla F$ maps $\text{int}(K) \rightarrow \text{int}(K^*)$ for any LHSCB).

By Euler's theorem for ν -log-homogeneous functions, $\langle s, \nabla F(s) \rangle = -\nu$, so the complementarity is

$$\langle s, \lambda \rangle = \mu \nu = \frac{\nu}{k^2}.$$

3 KKT system

At parameter k (equivalently $\mu = 1/k^2$), the primal-dual Newton system is:

$$\begin{bmatrix} \mu H(\lambda) & A \\ A^\top & 0 \end{bmatrix} \begin{bmatrix} \delta \lambda \\ x \end{bmatrix} = \begin{bmatrix} s - b \\ c - A^\top \lambda \end{bmatrix}, \quad (3)$$

where $H(\lambda) = \nabla^2 F^*(\lambda)$ is the dual barrier Hessian. The variable x is the *full* primal (not an increment), and $\delta \lambda$ is the dual update. After solving:

- $s + \delta s = Ax + b$ (primal feasible by construction),
- $A^\top(\lambda + \delta \lambda) = c$ (dual feasible by construction),
- $\text{gap} = \langle s + \delta s, \lambda + \delta \lambda \rangle$.

3.1 Schur complement

Using the identity $H(\lambda)^{-1} = \mu^2 \nabla^2 F(s)$ (which holds when $\lambda = -\mu \nabla F(s)$), the Schur complement gives:

$$(A^\top \nabla^2 F(s) A) x = A^\top (\nabla^2 F(s)(s - b) - \nabla F(s)) - k^2 c. \quad (4)$$

3.2 Scaled formulation: $w = k \cdot s$, $z = k \cdot x$

Substituting $s = w/k$ and using (2):

$$\nabla^2 F(s) = k^2 \nabla^2 F(w), \quad \nabla F(s) = k \nabla F(w).$$

The Schur complement (4) becomes, after dividing by k :

$$(A^\top \nabla^2 F(w) A) z = A^\top (\nabla^2 F(w) w - \nabla F(w)) - k(c + A^\top \nabla^2 F(w) b), \quad (5)$$

where $z = kx$. The Gram matrix $A^\top \nabla^2 F(w) A$ is **independent of k** .

3.3 Two-solve decomposition

Factor the Gram once, then two back-solves:

$$(A^\top \nabla^2 F(w) A) z_0 = A^\top (\nabla^2 F(w) w - \nabla F(w)) \quad (\text{centering}), \quad (6)$$

$$(A^\top \nabla^2 F(w) A) z_1 = -(c + A^\top \nabla^2 F(w) b) \quad (\text{optimality}). \quad (7)$$

The combined solution is $z(k) = z_0 + k \cdot z_1$, i.e.,

$$x(k) = \frac{z(k)}{k} = \frac{z_0}{k} + z_1.$$

Both c and b appear in (7), scaled by k symmetrically.

4 Line search for k

Define the cone-space directions $d_0 = Az_0$ and $d_1 = Az_1$. The displacement in w -space is $\delta w(k) = d_0 + k \cdot d_1$.

Dikin bound. The largest k with $\|\delta w(k)\|_{H(w)} \leq 1$ is the root of the quadratic

$$\langle d_1, d_1 \rangle_H k^2 + 2\langle d_0, d_1 \rangle_H k + (\langle d_0, d_0 \rangle_H - 1) = 0,$$

where $\langle a, b \rangle_H = a^\top \nabla^2 F(w) b$.

Primal-dual feasibility. Beyond the Dikin bound, binary search for the largest k such that:

$$1. \ s_{\text{new}} = Ax(k) + b \in \text{int}(K) \quad (\text{primal feasibility}),$$

$$2. \ \lambda_{\text{new}} = \lambda + \delta \lambda \in \text{int}(K^*) \quad (\text{dual feasibility}).$$

For low-dimensional cones (e.g., the 3D exponential cone), each feasibility check is $O(1)$. The geodesic step preserves feasibility beyond the Dikin ellipsoid.